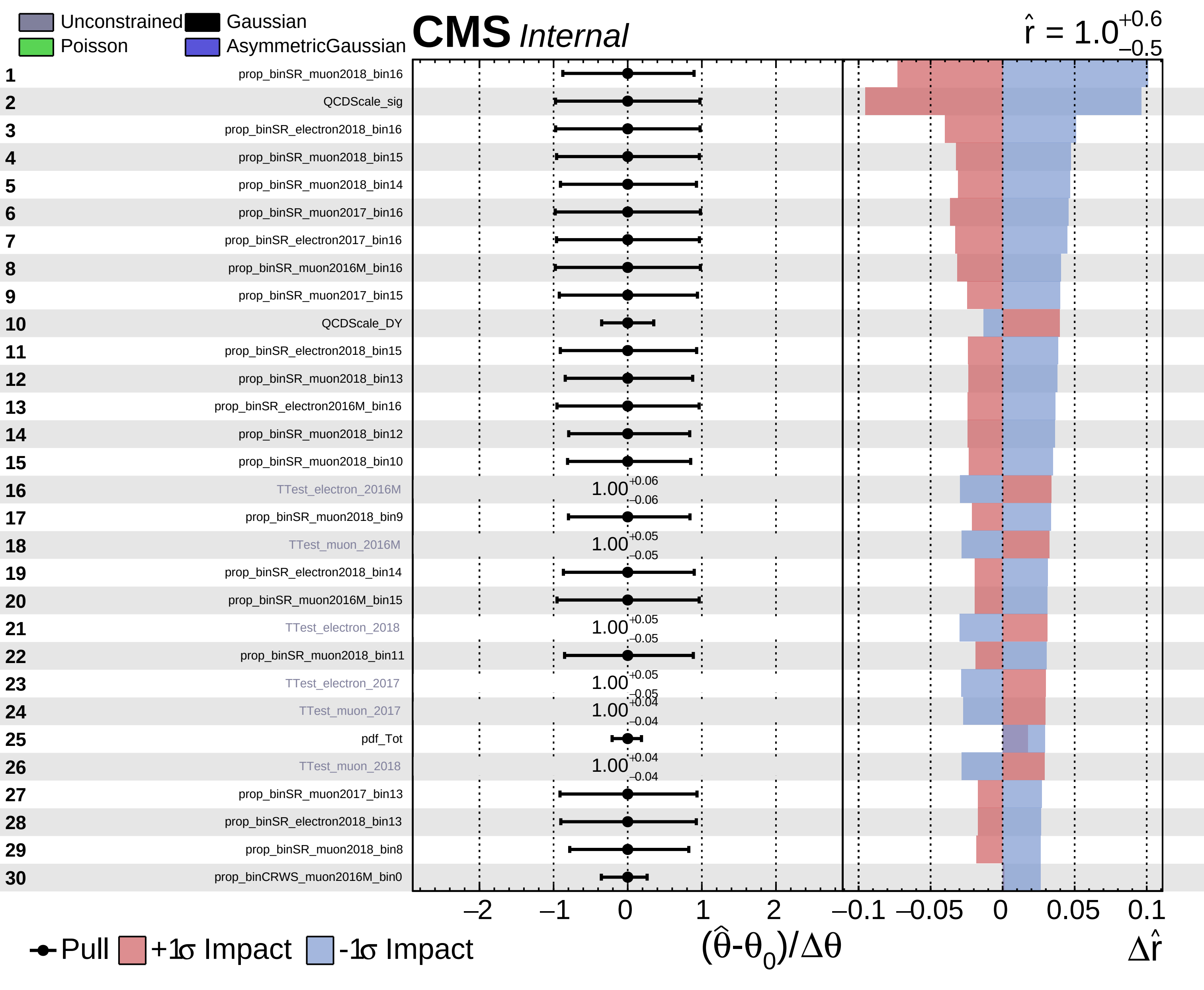
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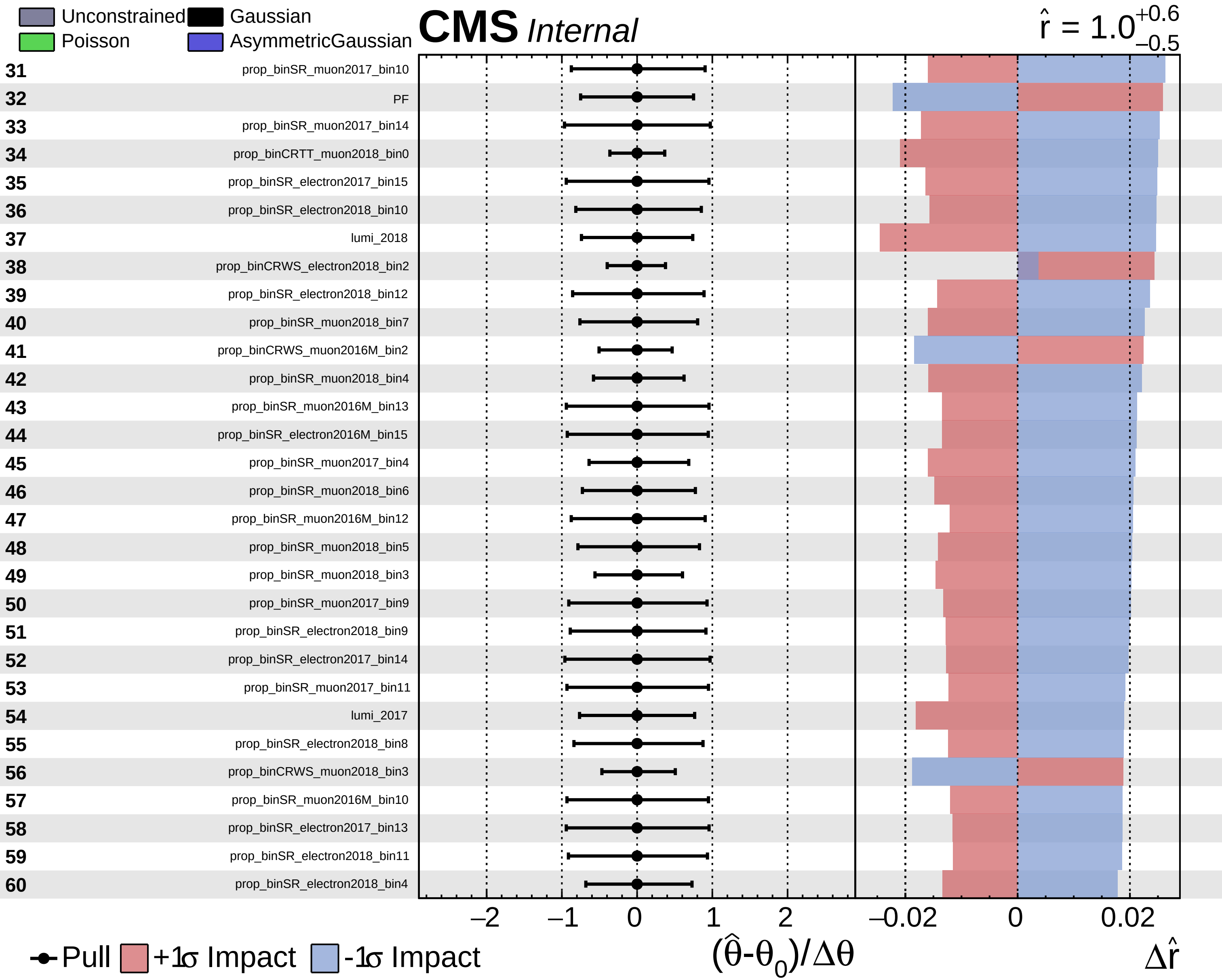
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
1 prop_binSR_muon2018_bin16
2 QCDScale_sig
3 prop_binSR_electron2018_bin16
4 prop_binSR_muon2018_bin15
5 prop_binSR_muon2018_bin14
6 prop_binSR_muon2017_bin16
7 prop_binSR_electron2017_bin16
8 prop_binSR_muon2016M_bin16
9 prop_binSR_muon2017_bin15
10 QCDScale_DY
11 prop_binSR_electron2018_bin15
12 prop_binSR_muon2018_bin13
13 prop_binSR_electron2016M_bin16
14 prop_binSR_muon2018_bin12
15 prop_binSR_muon2018_bin10
16 TTest_electron_2016M $1.00^{+0.06}_{-0.06}$
17 prop_binSR_muon2018_bin9
18 TTest_muon_2016M $1.00^{+0.05}_{-0.05}$
19 prop_binSR_electron2018_bin14
20 prop_binSR_muon2016M_bin15
21 TTest_electron_2018 $1.00^{+0.05}_{-0.05}$
22 prop_binSR_muon2018_bin11
23 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
24 TTest_muon_2017 $1.00^{+0.04}_{-0.04}$
25 pdf_Tot
26 TTest_muon_2018 $1.00^{+0.04}_{-0.04}$
27 prop_binSR_muon2017_bin13
28 prop_binSR_electron2018_bin13
29 prop_binSR_muon2018_bin8
30 prop_binCRWS_muon2016M_bin0
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$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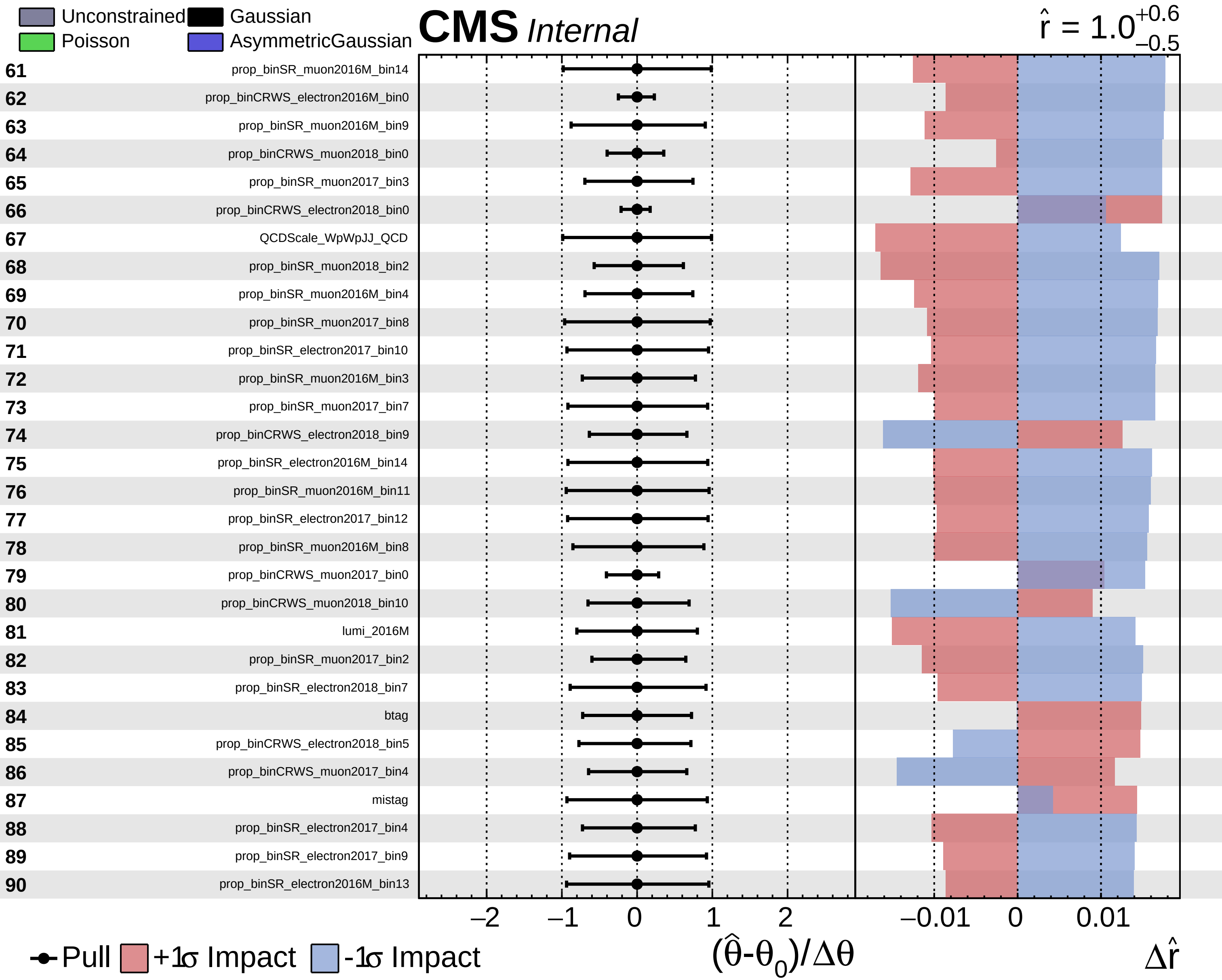
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
31 prop_binSR_muon2017_bin10
32 PF
33 prop_binSR_muon2017_bin14
34 prop_binCRTT_muon2018_bin0
35 prop_binSR_electron2017_bin15
36 prop_binSR_electron2018_bin10
37 lumi_2018
38 prop_binCRWS_electron2018_bin2
39 prop_binSR_electron2018_bin12
40 prop_binSR_muon2018_bin7
41 prop_binCRWS_muon2016M_bin2
42 prop_binSR_muon2018_bin4
43 prop_binSR_muon2016M_bin13
44 prop_binSR_electron2016M_bin15
45 prop_binSR_muon2017_bin4
46 prop_binSR_muon2018_bin6
47 prop_binSR_muon2016M_bin12
48 prop_binSR_muon2018_bin5
49 prop_binSR_muon2018_bin3
50 prop_binSR_muon2017_bin9
51 prop_binSR_electron2018_bin9
52 prop_binSR_electron2017_bin14
53 prop_binSR_muon2017_bin11
54 lumi_2017
55 prop_binSR_electron2018_bin8
56 prop_binCRWS_muon2018_bin3
57 prop_binSR_muon2016M_bin10
58 prop_binSR_electron2017_bin13
59 prop_binSR_electron2018_bin11
60 prop_binSR_electron2018_bin4
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



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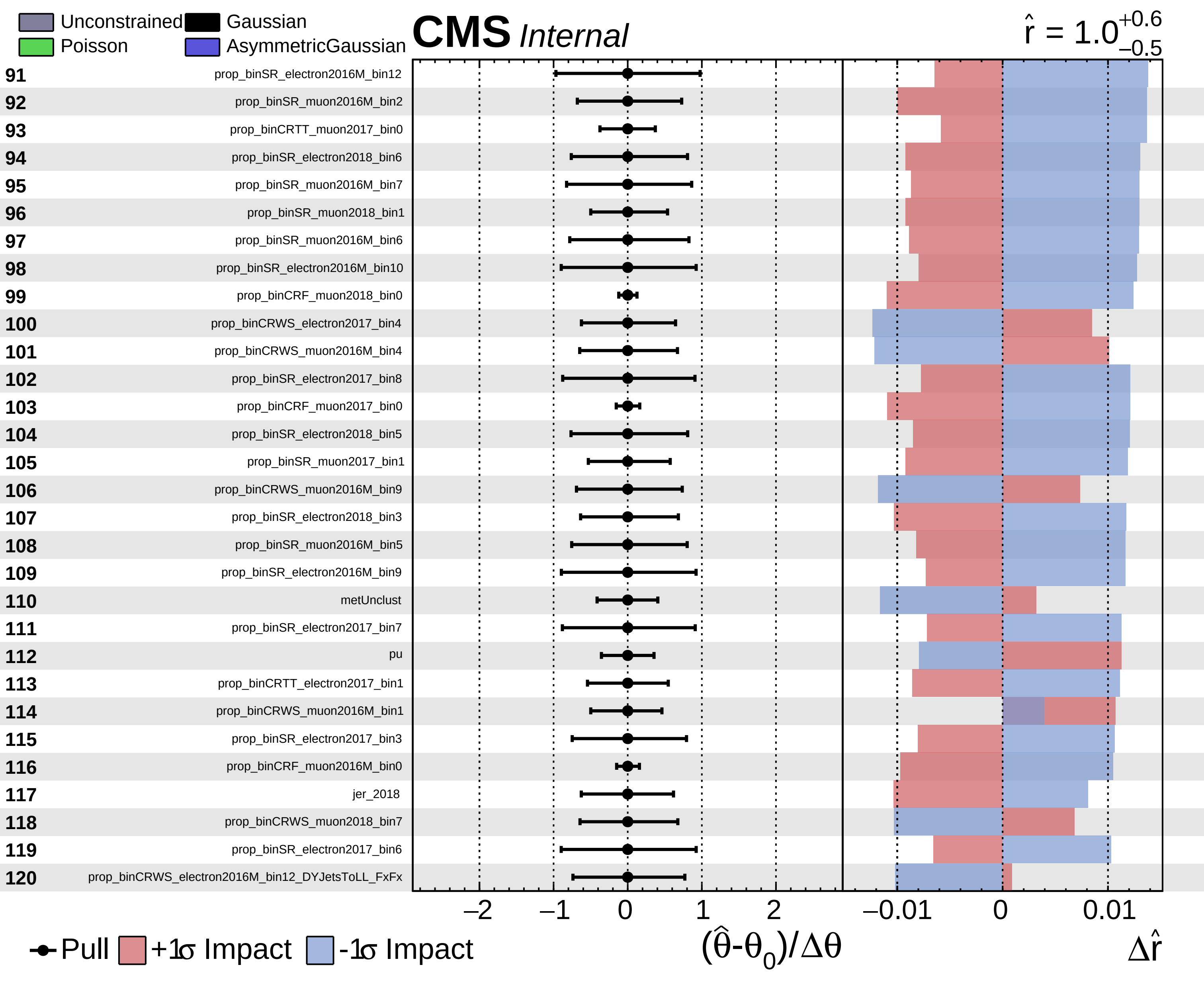
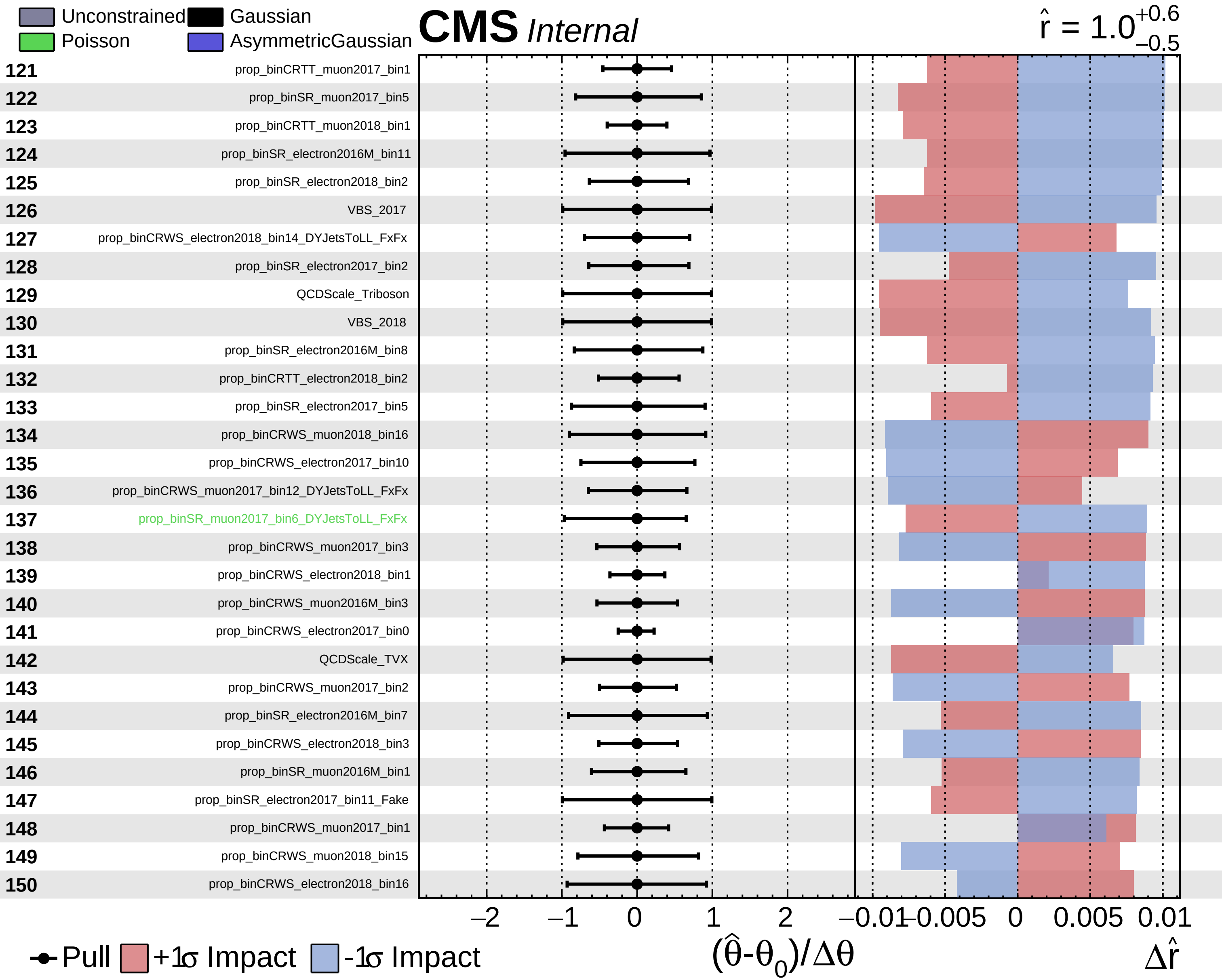
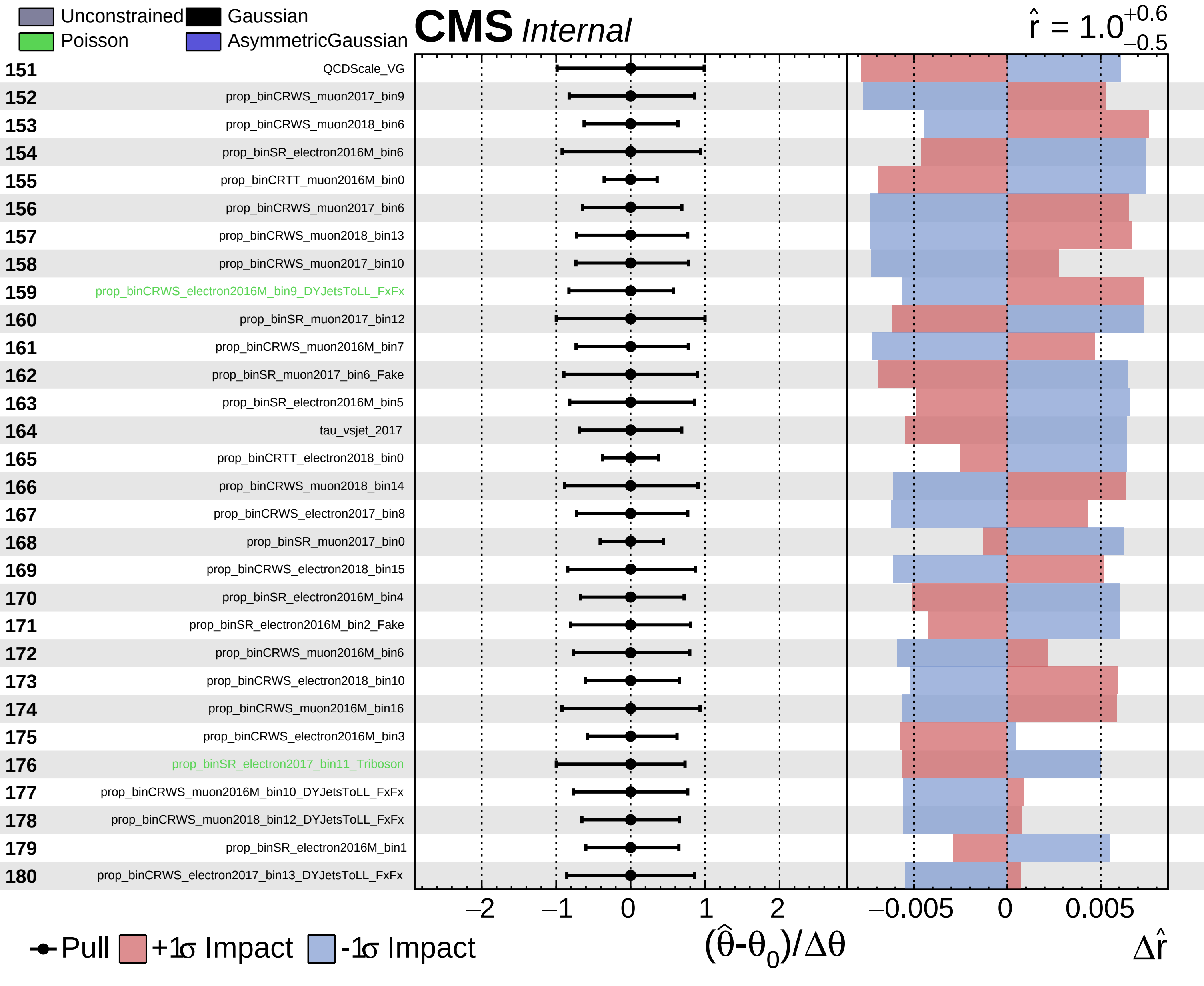
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
61 prop_binSR_muon2016M_bin14
62 prop_binCRWS_electron2016M_bin0
63 prop_binSR_muon2016M_bin9
64 prop_binCRWS_muon2018_bin0
65 prop_binSR_muon2017_bin3
66 prop_binCRWS_electron2018_bin0
67 QCDScale_WpWpJJ_QCD
68 prop_binSR_muon2018_bin2
69 prop_binSR_muon2016M_bin4
70 prop_binSR_muon2017_bin8
71 prop_binSR_electron2017_bin10
72 prop_binSR_muon2016M_bin3
73 prop_binSR_muon2017_bin7
74 prop_binCRWS_electron2018_bin9
75 prop_binSR_electron2016M_bin14
76 prop_binSR_muon2016M_bin11
77 prop_binSR_electron2017_bin12
78 prop_binSR_muon2016M_bin8
79 prop_binCRWS_muon2017_bin0
80 prop_binCRWS_muon2018_bin10
81 lumi_2016M
82 prop_binSR_muon2017_bin2
83 prop_binSR_electron2018_bin7
84 btag
85 prop_binCRWS_electron2018_bin5
86 prop_binCRWS_muon2017_bin4
87 mistag
88 prop_binSR_electron2017_bin4
89 prop_binSR_electron2017_bin9
90 prop_binSR_electron2016M_bin13
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

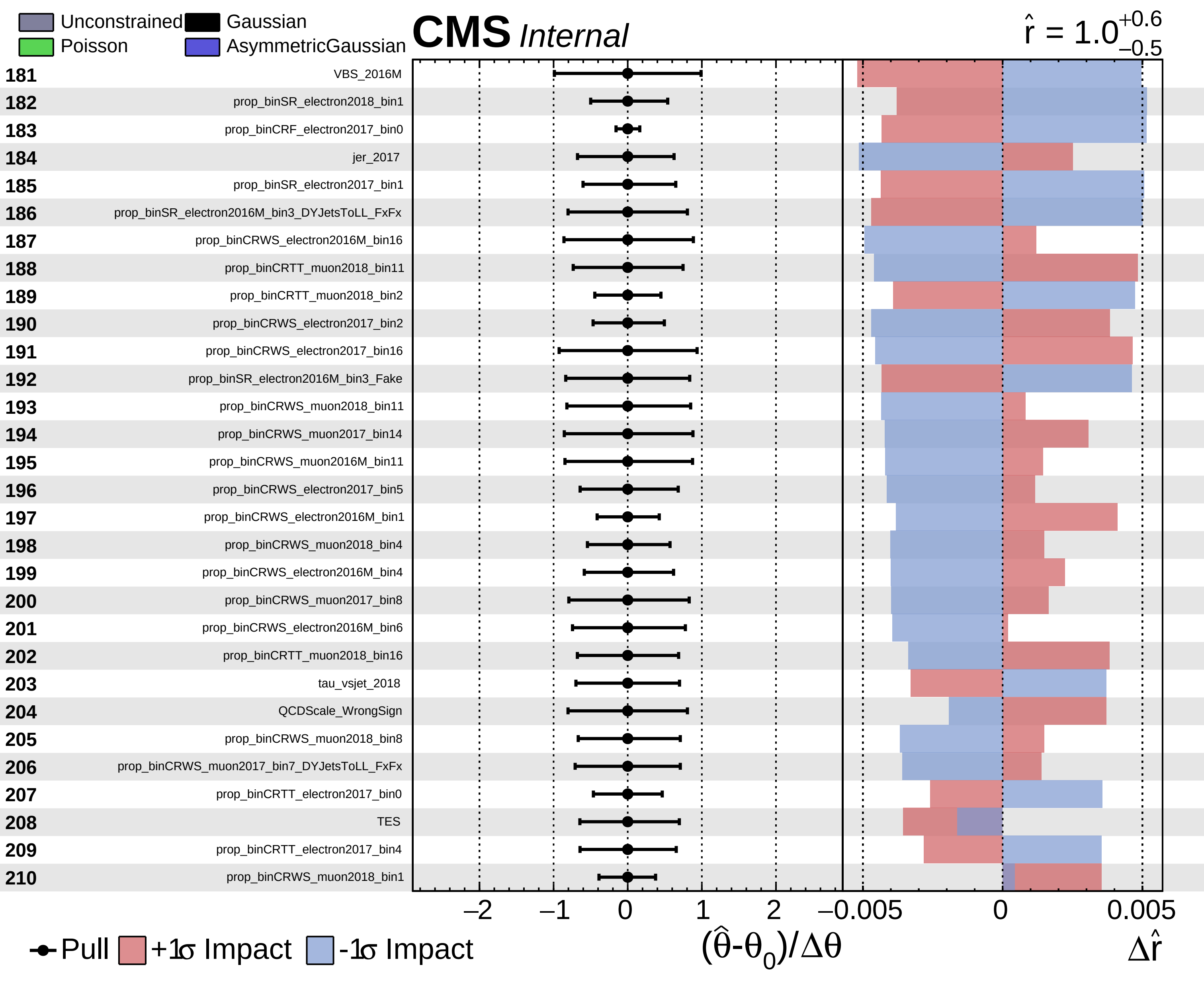


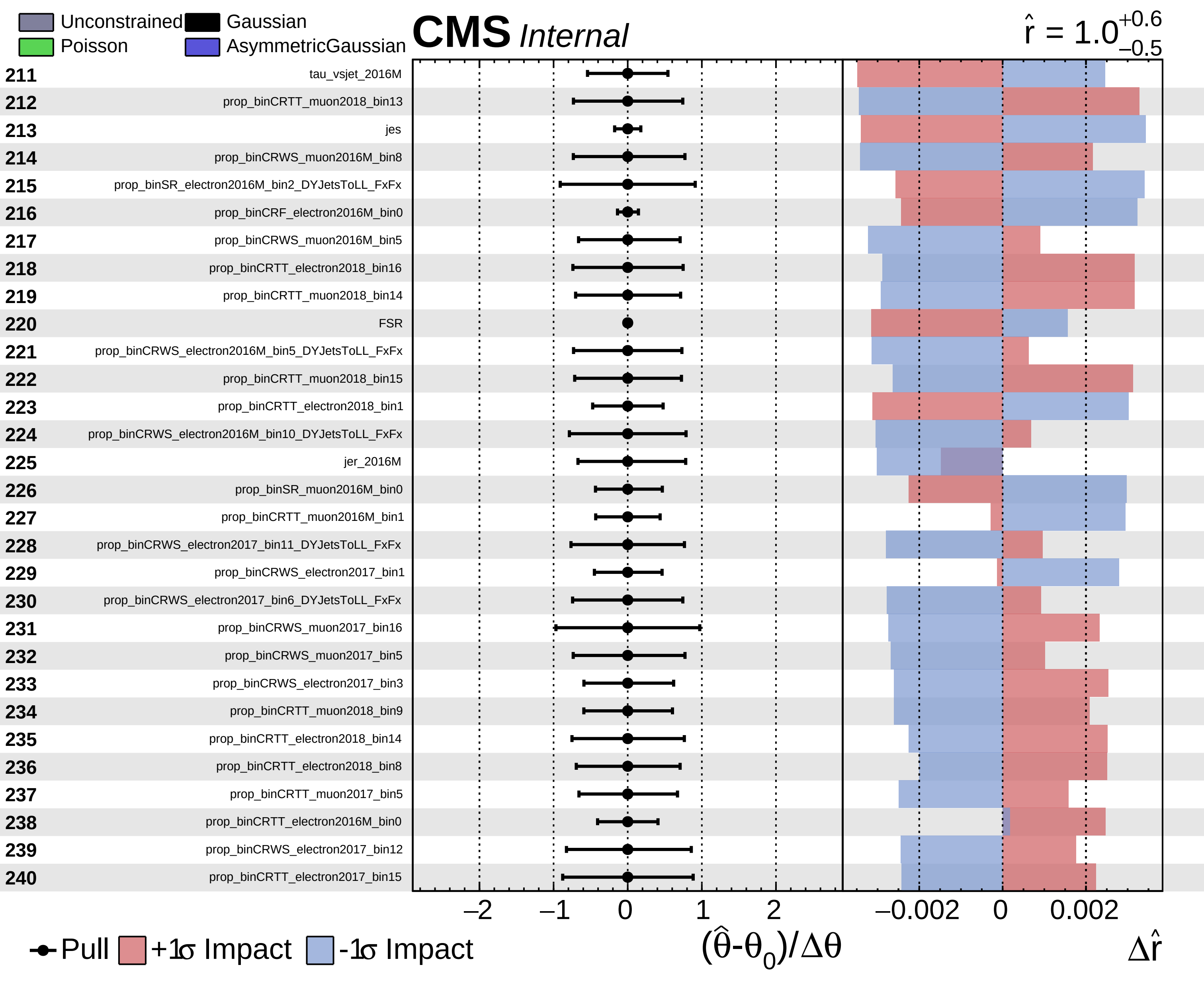
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
91 prop_binSR_electron2016M_bin12
92 prop_binSR_muon2016M_bin2
93 prop_binCRTT_muon2017_bin0
94 prop_binSR_electron2018_bin6
95 prop_binSR_muon2016M_bin7
96 prop_binSR_muon2018_bin1
97 prop_binSR_muon2016M_bin6
98 prop_binSR_electron2016M_bin10
99 prop_binCRF_muon2018_bin0
100 prop_binCRWS_electron2017_bin4
101 prop_binCRWS_muon2016M_bin4
102 prop_binSR_electron2017_bin8
103 prop_binCRF_muon2017_bin0
104 prop_binSR_electron2018_bin5
105 prop_binSR_muon2017_bin1
106 prop_binCRWS_muon2016M_bin9
107 prop_binSR_electron2018_bin3
108 prop_binSR_muon2016M_bin5
109 prop_binSR_electron2016M_bin9
110 metUnclust
111 prop_binSR_electron2017_bin7
112 pu
113 prop_binCRTT_electron2017_bin1
114 prop_binCRWS_muon2016M_bin1
115 prop_binSR_electron2017_bin3
116 prop_binCRF_muon2016M_bin0
117 jer_2018
118 prop_binCRWS_muon2018_bin7
119 prop_binSR_electron2017_bin6
120 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
121 prop_binCRTT_muon2017_bin1
122 prop_binSR_muon2017_bin5
123 prop_binCRTT_muon2018_bin1
124 prop_binSR_electron2016M_bin11
125 prop_binSR_electron2018_bin2
126 VBS_2017
127 prop_binCRWS_electron2018_bin14_DYJetsToLL_FxFx
128 prop_binSR_electron2017_bin2
129 QCDScale_Triboson
130 VBS_2018
131 prop_binSR_electron2016M_bin8
132 prop_binCRTT_electron2018_bin2
133 prop_binSR_electron2017_bin5
134 prop_binCRWS_muon2018_bin16
135 prop_binCRWS_electron2017_bin10
136 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
137 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
138 prop_binCRWS_muon2017_bin3
139 prop_binCRWS_electron2018_bin1
140 prop_binCRWS_muon2016M_bin3
141 prop_binCRWS_electron2017_bin0
142 QCDScale_TVX
143 prop_binCRWS_muon2017_bin2
144 prop_binSR_electron2016M_bin7
145 prop_binCRWS_electron2018_bin3
146 prop_binSR_muon2016M_bin1
147 prop_binSR_electron2017_bin11_Fake
148 prop_binCRWS_muon2017_bin1
149 prop_binCRWS_muon2018_bin15
150 prop_binCRWS_electron2018_bin16
−2
−1
0
1
2
−0.01
−0.005
0
0.005
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

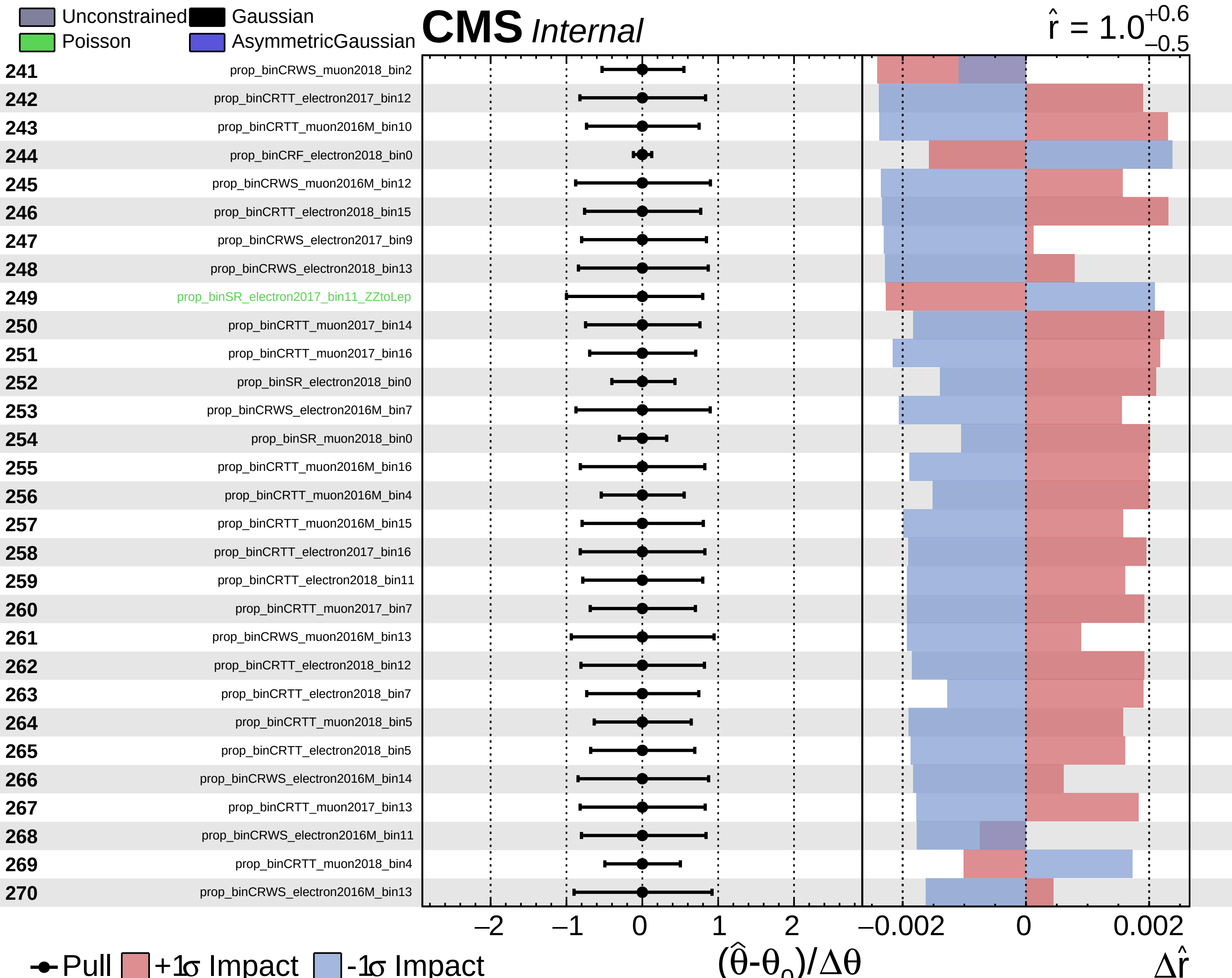


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
151 QCDScale_VG
152 prop_binCRWS_muon2017_bin9
153 prop_binCRWS_muon2018_bin6
154 prop_binSR_electron2016M_bin6
155 prop_binCRTT_muon2016M_bin0
156 prop_binCRWS_muon2017_bin6
157 prop_binCRWS_muon2018_bin13
158 prop_binCRWS_muon2017_bin10
159 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
160 prop_binSR_muon2017_bin12
161 prop_binCRWS_muon2016M_bin7
162 prop_binSR_muon2017_bin6_Fake
163 prop_binSR_electron2016M_bin5
164 tau_vsjet_2017
165 prop_binCRTT_electron2018_bin0
166 prop_binCRWS_muon2018_bin14
167 prop_binCRWS_electron2017_bin8
168 prop_binSR_muon2017_bin0
169 prop_binCRWS_electron2018_bin15
170 prop_binSR_electron2016M_bin4
171 prop_binSR_electron2016M_bin2_Fake
172 prop_binCRWS_muon2016M_bin6
173 prop_binCRWS_electron2018_bin10
174 prop_binCRWS_muon2016M_bin16
175 prop_binCRWS_electron2016M_bin3
176 prop_binSR_electron2017_bin11_Triboson
177 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
178 prop_binCRWS_muon2018_bin12_DYJetsToLL_FxFx
179 prop_binSR_electron2016M_bin1
180 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
181 VBS_2016M
182 prop_binSR_electron2018_bin1
183 prop_binCRF_electron2017_bin0
184 jer_2017
185 prop_binSR_electron2017_bin1
186 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
187 prop_binCRWS_electron2016M_bin16
188 prop_binCRTT_muon2018_bin11
189 prop_binCRTT_muon2018_bin2
190 prop_binCRWS_electron2017_bin2
191 prop_binCRWS_electron2017_bin16
192 prop_binSR_electron2016M_bin3_Fake
193 prop_binCRWS_muon2018_bin11
194 prop_binCRWS_muon2017_bin14
195 prop_binCRWS_muon2016M_bin11
196 prop_binCRWS_electron2017_bin5
197 prop_binCRWS_electron2016M_bin1
198 prop_binCRWS_muon2018_bin4
199 prop_binCRWS_electron2016M_bin4
200 prop_binCRWS_muon2017_bin8
201 prop_binCRWS_electron2016M_bin6
202 prop_binCRTT_muon2018_bin16
203 tau_vsjet_2018
204 QCDScale_WrongSign
205 prop_binCRWS_muon2018_bin8
206 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
207 prop_binCRTT_electron2017_bin0
208 TES
209 prop_binCRTT_electron2017_bin4
210 prop_binCRWS_muon2018_bin1
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
211 tau_vsjet_2016M
212 prop_binCRTT_muon2018_bin13
213 jes
214 prop_binCRWS_muon2016M_bin8
215 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
216 prop_binCRF_electron2016M_bin0
217 prop_binCRWS_muon2016M_bin5
218 prop_binCRTT_electron2018_bin16
219 prop_binCRTT_muon2018_bin14
220 FSR
221 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
222 prop_binCRTT_muon2018_bin15
223 prop_binCRTT_electron2018_bin1
224 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
225 jer_2016M
226 prop_binSR_muon2016M_bin0
227 prop_binCRTT_muon2016M_bin1
228 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
229 prop_binCRWS_electron2017_bin1
230 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
231 prop_binCRWS_muon2017_bin16
232 prop_binCRWS_muon2017_bin5
233 prop_binCRWS_electron2017_bin3
234 prop_binCRTT_muon2018_bin9
235 prop_binCRTT_electron2018_bin14
236 prop_binCRTT_electron2018_bin8
237 prop_binCRTT_muon2017_bin5
238 prop_binCRTT_electron2016M_bin0
239 prop_binCRWS_electron2017_bin12
240 prop_binCRTT_electron2017_bin15
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
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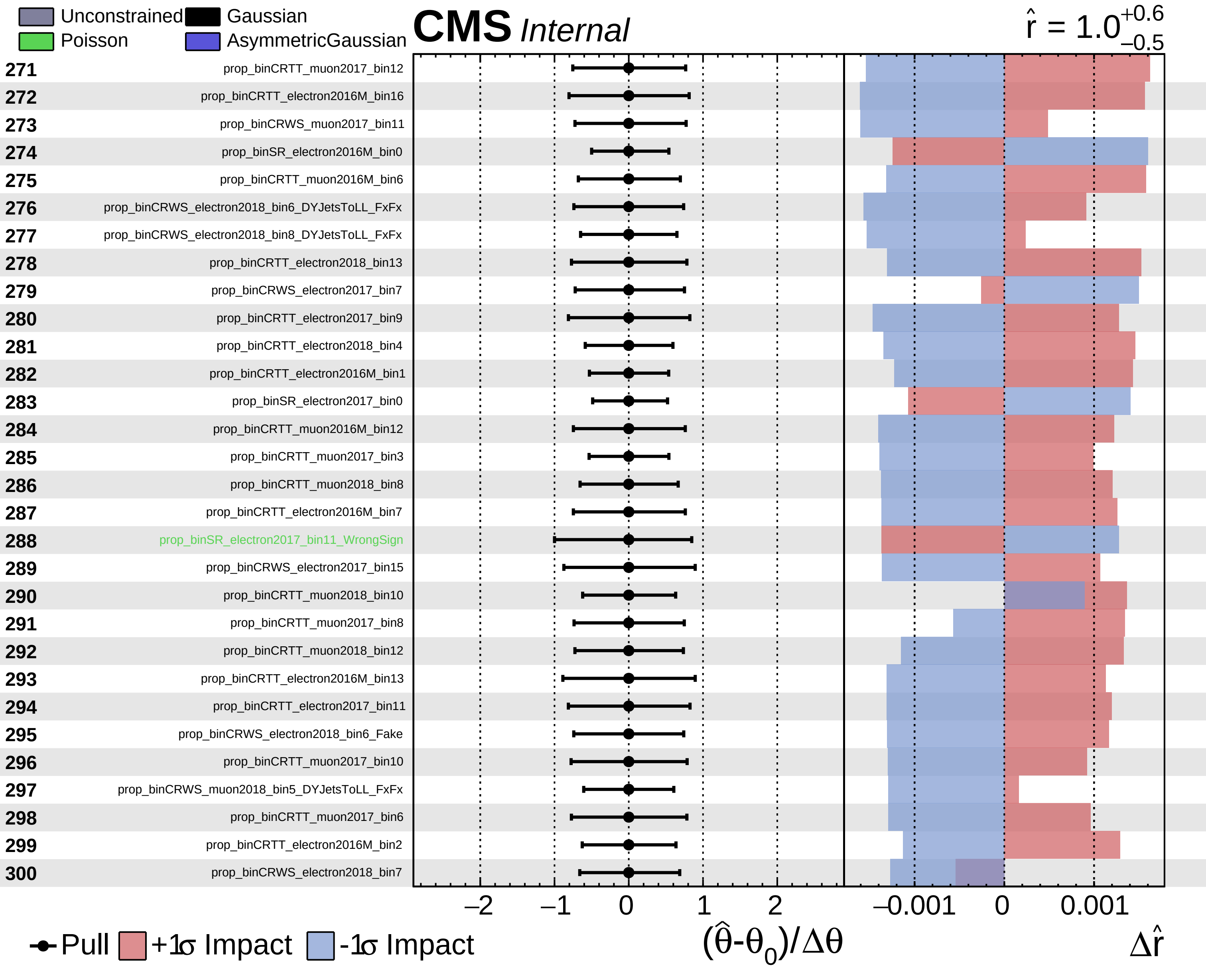
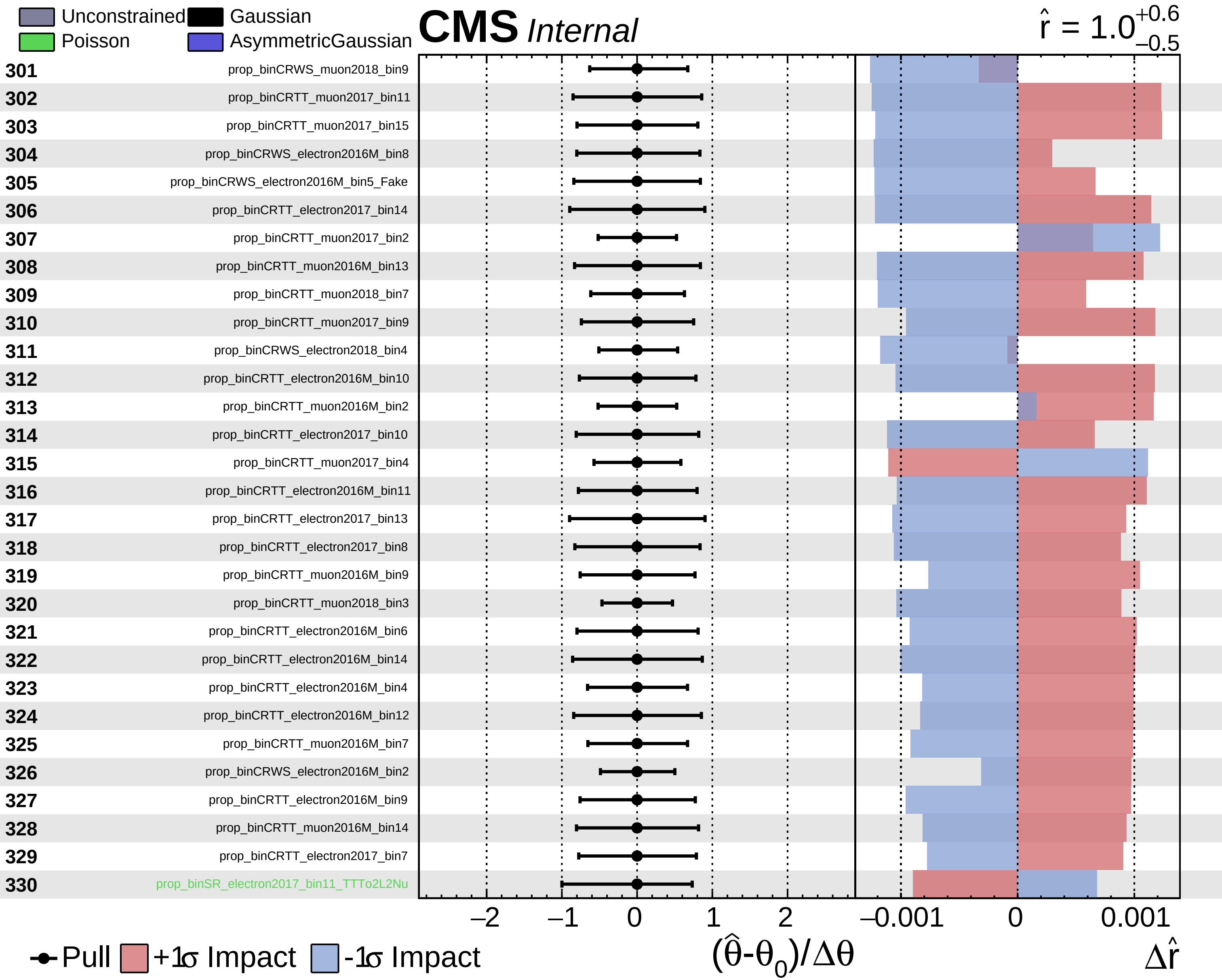
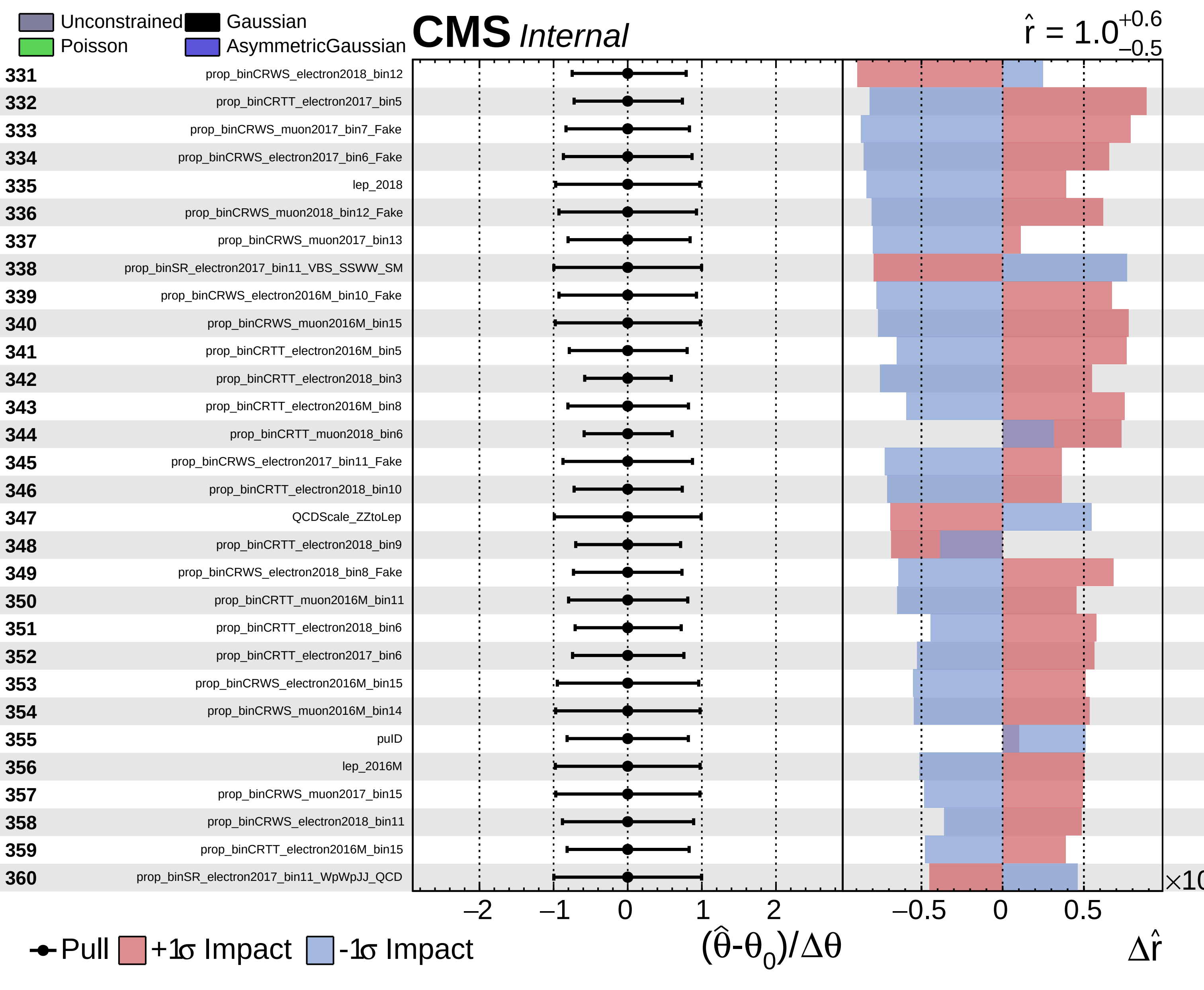
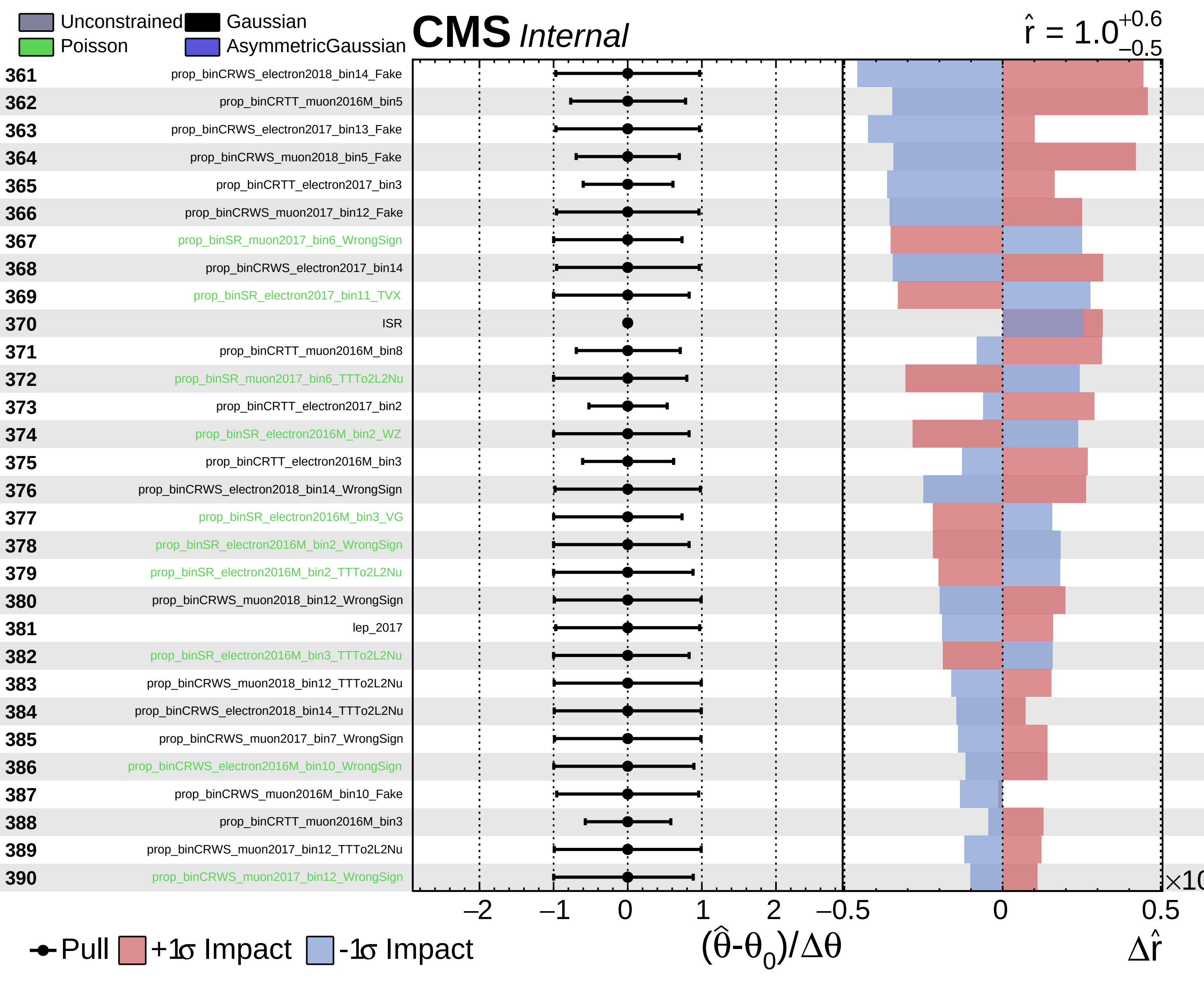
242 prop_binCRTT_electron2017_bin12
243 prop_binCRTT_muon2016M_bin10
244 prop_binCRF_electron2018_bin0
245 prop_binCRWS_muon2016M_bin12
246 prop_binCRTT_electron2018_bin15
247 prop_binCRWS_electron2017_bin9
248 prop_binCRWS_electron2018_bin13
249 prop_binSR_electron2017_bin11_ZZtoLep
250 prop_binCRTT_muon2017_bin14
251 prop_binCRTT_muon2017_bin16
252 prop_binSR_electron2018_bin0
253 prop_binCRWS_electron2016M_bin7
254 prop_binSR_muon2018_bin0
255 prop_binCRTT_muon2016M_bin16
256 prop_binCRTT_muon2016M_bin4
257 prop_binCRTT_muon2016M_bin15
258 prop_binCRTT_electron2017_bin16
259 prop_binCRTT_electron2018_bin11
260 prop_binCRTT_muon2017_bin7
261 prop_binCRWS_muon2016M_bin13
262 prop_binCRTT_electron2018_bin12
263 prop_binCRTT_electron2018_bin7
264 prop_binCRTT_muon2018_bin5
265 prop_binCRTT_electron2018_bin5
266 prop_binCRWS_electron2016M_bin14
267 prop_binCRTT_muon2017_bin13
268 prop_binCRWS_electron2016M_bin11
269 prop_binCRTT_muon2018_bin4
270 prop_binCRWS_electron2016M_bin13
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



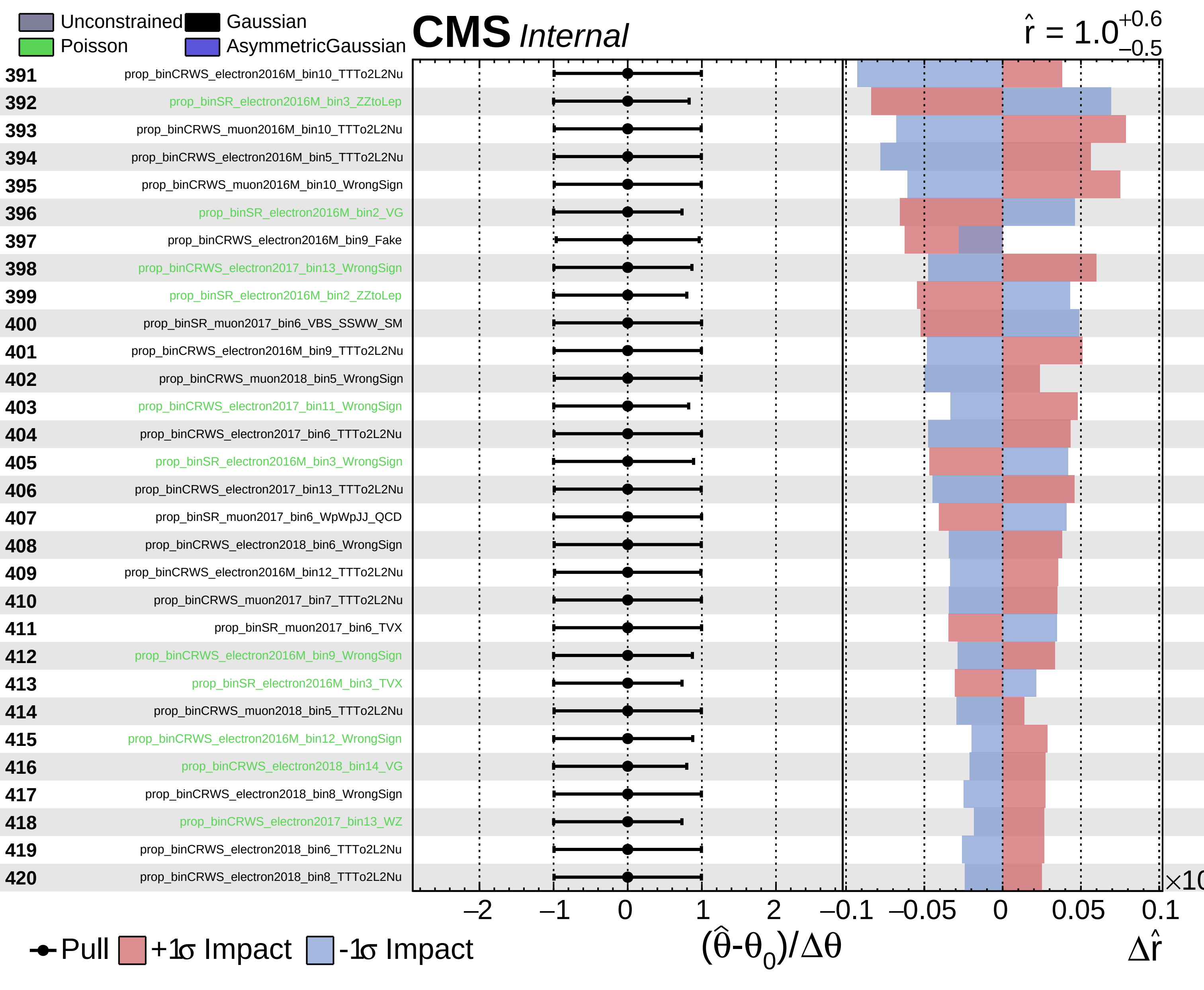
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
271 prop_binCRTT_muon2017_bin12
272 prop_binCRTT_electron2016M_bin16
273 prop_binCRWS_muon2017_bin11
274 prop_binSR_electron2016M_bin0
275 prop_binCRTT_muon2016M_bin6
276 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
277 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
278 prop_binCRTT_electron2018_bin13
279 prop_binCRWS_electron2017_bin7
280 prop_binCRTT_electron2017_bin9
281 prop_binCRTT_electron2018_bin4
282 prop_binCRTT_electron2016M_bin1
283 prop_binSR_electron2017_bin0
284 prop_binCRTT_muon2016M_bin12
285 prop_binCRTT_muon2017_bin3
286 prop_binCRTT_muon2018_bin8
287 prop_binCRTT_electron2016M_bin7
288 prop_binSR_electron2017_bin11_WrongSign
289 prop_binCRWS_electron2017_bin15
290 prop_binCRTT_muon2018_bin10
291 prop_binCRTT_muon2017_bin8
292 prop_binCRTT_muon2018_bin12
293 prop_binCRTT_electron2016M_bin13
294 prop_binCRTT_electron2017_bin11
295 prop_binCRWS_electron2018_bin6_Fake
296 prop_binCRTT_muon2017_bin10
297 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
298 prop_binCRTT_muon2017_bin6
299 prop_binCRTT_electron2016M_bin2
300 prop_binCRWS_electron2018_bin7
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
301 prop_binCRWS_muon2018_bin9
302 prop_binCRTT_muon2017_bin11
303 prop_binCRTT_muon2017_bin15
304 prop_binCRWS_electron2016M_bin8
305 prop_binCRWS_electron2016M_bin5_Fake
306 prop_binCRTT_electron2017_bin14
307 prop_binCRTT_muon2017_bin2
308 prop_binCRTT_muon2016M_bin13
309 prop_binCRTT_muon2018_bin7
310 prop_binCRTT_muon2017_bin9
311 prop_binCRWS_electron2018_bin4
312 prop_binCRTT_electron2016M_bin10
313 prop_binCRTT_muon2016M_bin2
314 prop_binCRTT_electron2017_bin10
315 prop_binCRTT_muon2017_bin4
316 prop_binCRTT_electron2016M_bin11
317 prop_binCRTT_electron2017_bin13
318 prop_binCRTT_electron2017_bin8
319 prop_binCRTT_muon2016M_bin9
320 prop_binCRTT_muon2018_bin3
321 prop_binCRTT_electron2016M_bin6
322 prop_binCRTT_electron2016M_bin14
323 prop_binCRTT_electron2016M_bin4
324 prop_binCRTT_electron2016M_bin12
325 prop_binCRTT_muon2016M_bin7
326 prop_binCRWS_electron2016M_bin2
327 prop_binCRTT_electron2016M_bin9
328 prop_binCRTT_muon2016M_bin14
329 prop_binCRTT_electron2017_bin7
330 prop_binSR_electron2017_bin11_TTTo2L2Nu
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

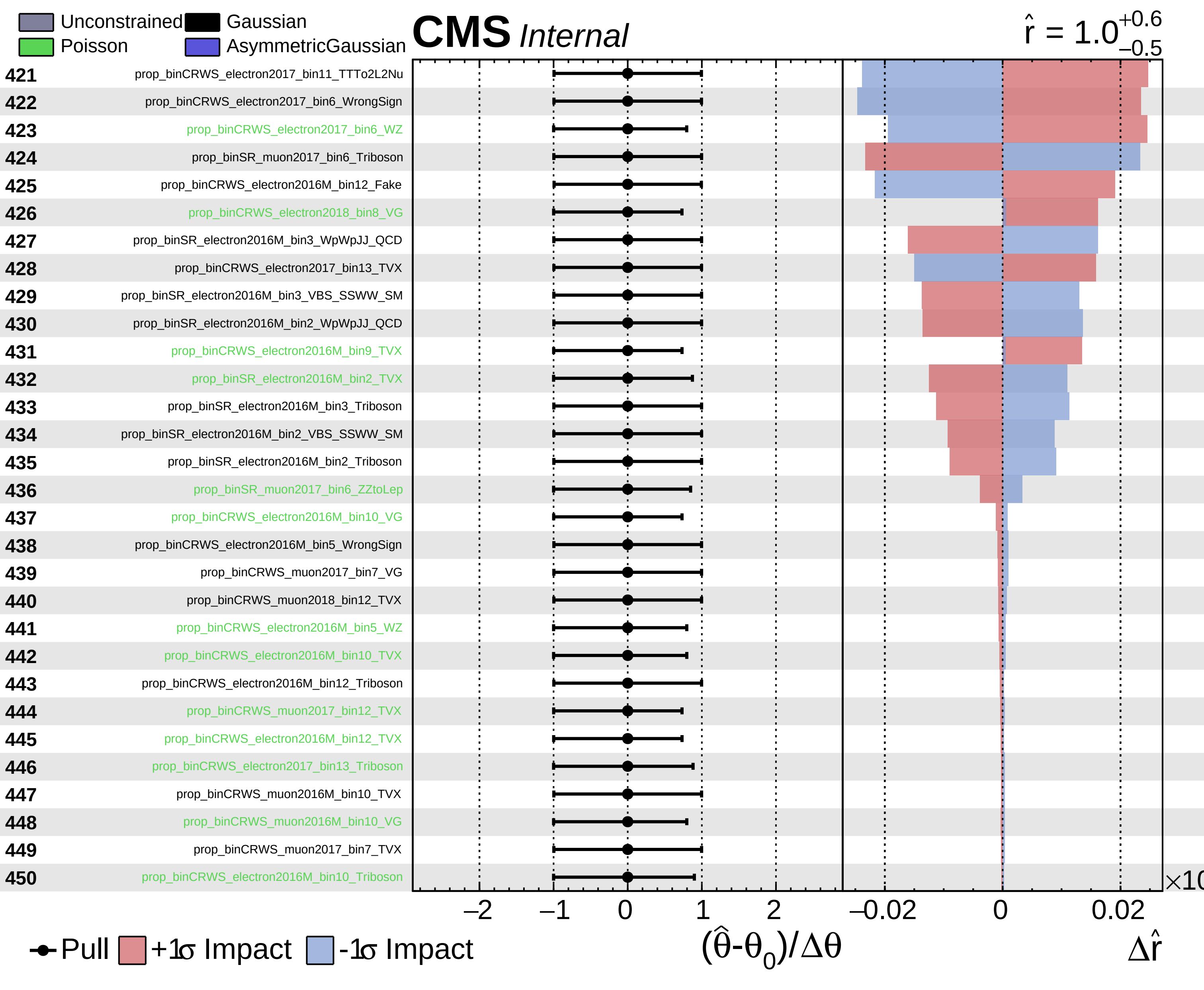

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
331 prop_binCRWS_electron2018_bin12
332 prop_binCRTT_electron2017_bin5
333 prop_binCRWS_muon2017_bin7_Fake
334 prop_binCRWS_electron2017_bin6_Fake
335 lep_2018
336 prop_binCRWS_muon2018_bin12_Fake
337 prop_binCRWS_muon2017_bin13
338 prop_binSR_electron2017_bin11_VBS_SSWW_SM
339 prop_binCRWS_electron2016M_bin10_Fake
340 prop_binCRWS_muon2016M_bin15
341 prop_binCRTT_electron2016M_bin5
342 prop_binCRTT_electron2018_bin3
343 prop_binCRTT_electron2016M_bin8
344 prop_binCRTT_muon2018_bin6
345 prop_binCRWS_electron2017_bin11_Fake
346 prop_binCRTT_electron2018_bin10
347 QCDScale_ZZtoLep
348 prop_binCRTT_electron2018_bin9
349 prop_binCRWS_electron2018_bin8_Fake
350 prop_binCRTT_muon2016M_bin11
351 prop_binCRTT_electron2018_bin6
352 prop_binCRTT_electron2017_bin6
353 prop_binCRWS_electron2016M_bin15
354 prop_binCRWS_muon2016M_bin14
355 puID
356 lep_2016M
357 prop_binCRWS_muon2017_bin15
358 prop_binCRWS_electron2018_bin11
359 prop_binCRTT_electron2016M_bin15
360 prop_binSR_electron2017_bin11_WpWpJJ_QCD
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



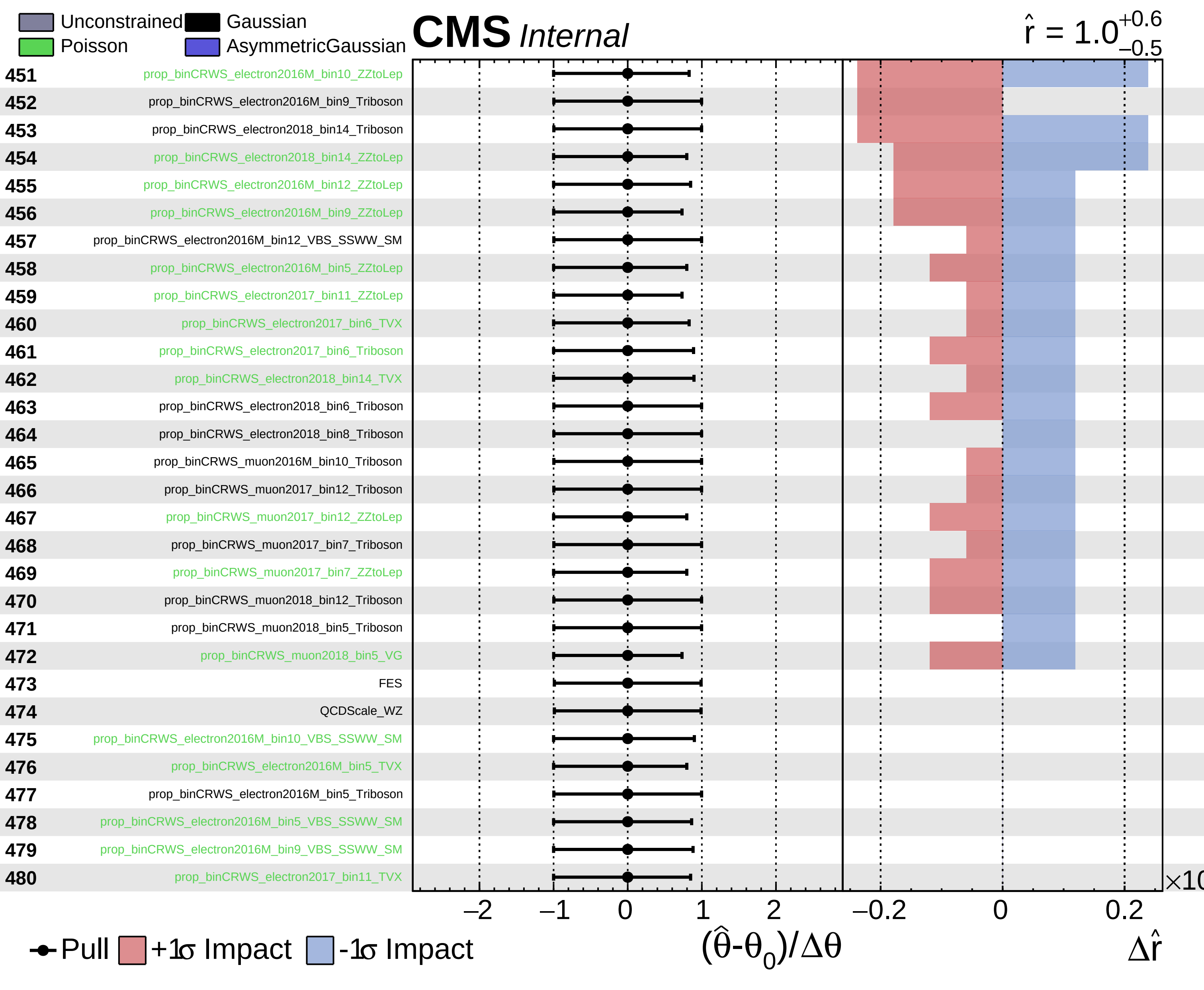
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
361 prop_binCRWS_electron2018_bin14_Fake
362 prop_binCRTT_muon2016M_bin5
363 prop_binCRWS_electron2017_bin13_Fake
364 prop_binCRWS_muon2018_bin5_Fake
365 prop_binCRTT_electron2017_bin3
366 prop_binCRWS_muon2017_bin12_Fake
367 prop_binSR_muon2017_bin6_WrongSign
368 prop_binCRWS_electron2017_bin14
369 prop_binSR_electron2017_bin11_TVX
370 ISR
371 prop_binCRTT_muon2016M_bin8
372 prop_binSR_muon2017_bin6_TTTo2L2Nu
373 prop_binCRTT_electron2017_bin2
374 prop_binSR_electron2016M_bin2_WZ
375 prop_binCRTT_electron2016M_bin3
376 prop_binCRWS_electron2018_bin14_WrongSign
377 prop_binSR_electron2016M_bin3_VG
378 prop_binSR_electron2016M_bin2_WrongSign
379 prop_binSR_electron2016M_bin2_TTTo2L2Nu
380 prop_binCRWS_muon2018_bin12_WrongSign
381 lep_2017
382 prop_binSR_electron2016M_bin3_TTTo2L2Nu
383 prop_binCRWS_muon2018_bin12_TTTo2L2Nu
384 prop_binCRWS_electron2018_bin14_TTTo2L2Nu
385 prop_binCRWS_muon2017_bin7_WrongSign
386 prop_binCRWS_electron2016M_bin10_WrongSign
387 prop_binCRWS_muon2016M_bin10_Fake
388 prop_binCRTT_muon2016M_bin3
389 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
390 prop_binCRWS_muon2017_bin12_WrongSign
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



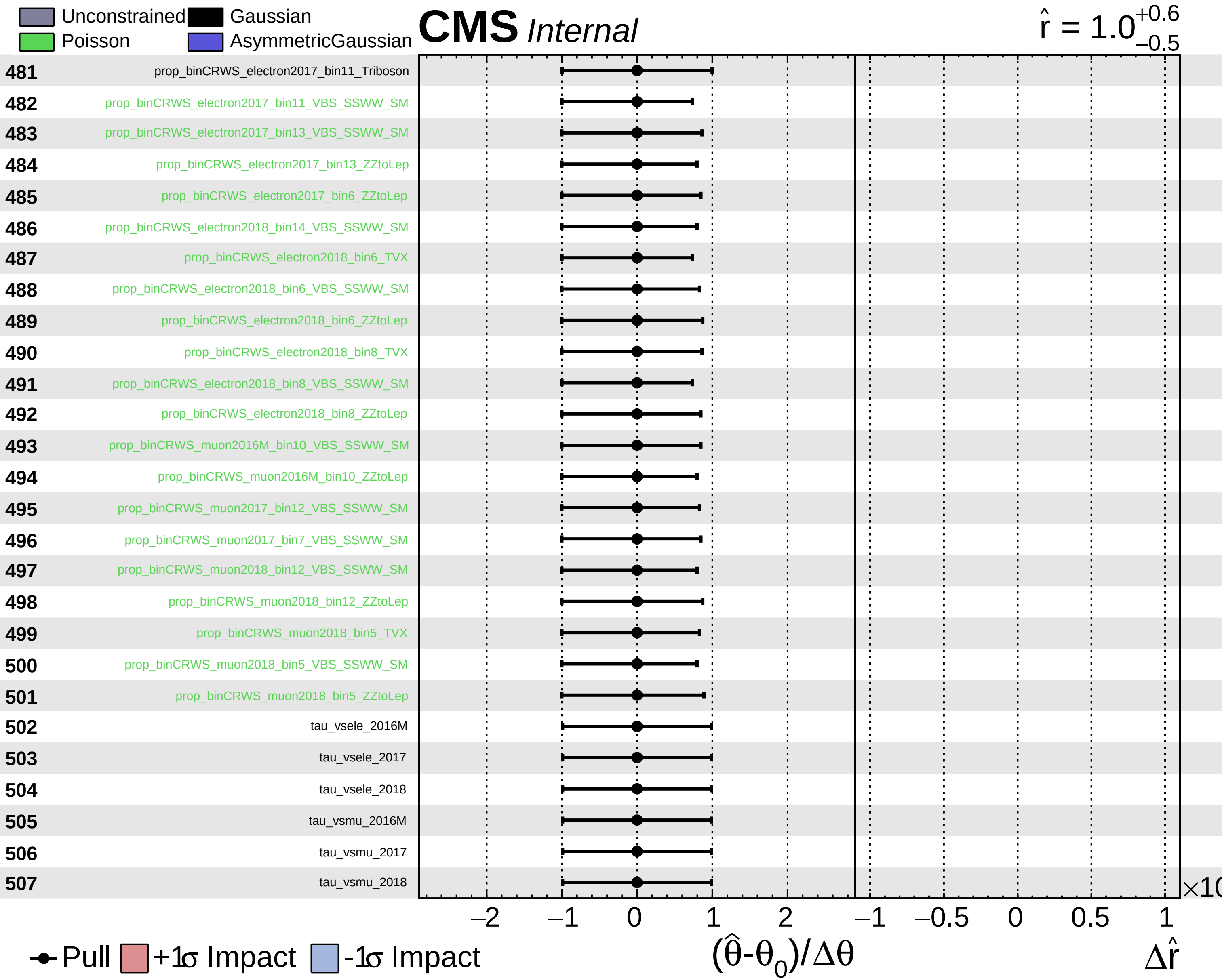
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
391 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
392 prop_binSR_electron2016M_bin3_ZZtoLep
393 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
394 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
395 prop_binCRWS_muon2016M_bin10_WrongSign
396 prop_binSR_electron2016M_bin2_VG
397 prop_binCRWS_electron2016M_bin9_Fake
398 prop_binCRWS_electron2017_bin13_WrongSign
399 prop_binSR_electron2016M_bin2_ZZtoLep
400 prop_binSR_muon2017_bin6_VBS_SSWW_SM
401 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
402 prop_binCRWS_muon2018_bin5_WrongSign
403 prop_binCRWS_electron2017_bin11_WrongSign
404 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
405 prop_binSR_electron2016M_bin3_WrongSign
406 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
407 prop_binSR_muon2017_bin6_WpWpJJ_QCD
408 prop_binCRWS_electron2018_bin6_WrongSign
409 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
410 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
411 prop_binSR_muon2017_bin6_TVX
412 prop_binCRWS_electron2016M_bin9_WrongSign
413 prop_binSR_electron2016M_bin3_TVX
414 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
415 prop_binCRWS_electron2016M_bin12_WrongSign
416 prop_binCRWS_electron2018_bin14_VG
417 prop_binCRWS_electron2018_bin8_WrongSign
418 prop_binCRWS_electron2017_bin13_WZ
419 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
420 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
421 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
422 prop_binCRWS_electron2017_bin6_WrongSign
423 prop_binCRWS_electron2017_bin6_WZ
424 prop_binSR_muon2017_bin6_Triboson
425 prop_binCRWS_electron2016M_bin12_Fake
426 prop_binCRWS_electron2018_bin8_VG
427 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
428 prop_binCRWS_electron2017_bin13_TVX
429 prop_binSR_electron2016M_bin3_VBS_SSWW_SM
430 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
431 prop_binCRWS_electron2016M_bin9_TVX
432 prop_binSR_electron2016M_bin2_TVX
433 prop_binSR_electron2016M_bin3_Triboson
434 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
435 prop_binSR_electron2016M_bin2_Triboson
436 prop_binSR_muon2017_bin6_ZZtoLep
437 prop_binCRWS_electron2016M_bin10_VG
438 prop_binCRWS_electron2016M_bin5_WrongSign
439 prop_binCRWS_muon2017_bin7_VG
440 prop_binCRWS_muon2018_bin12_TVX
441 prop_binCRWS_electron2016M_bin5_WZ
442 prop_binCRWS_electron2016M_bin10_TVX
443 prop_binCRWS_electron2016M_bin12_Triboson
444 prop_binCRWS_muon2017_bin12_TVX
445 prop_binCRWS_electron2016M_bin12_TVX
446 prop_binCRWS_electron2017_bin13_Triboson
447 prop_binCRWS_muon2016M_bin10_TVX
448 prop_binCRWS_muon2016M_bin10_VG
449 prop_binCRWS_muon2017_bin7_TVX
450 prop_binCRWS_electron2016M_bin10_Triboson
−2
−1
0
1
2
−0.02
0
0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0+0.6 −0.5
451 prop_binCRWS_electron2016M_bin10_ZZtoLep
452 prop_binCRWS_electron2016M_bin9_Triboson
453 prop_binCRWS_electron2018_bin14_Triboson
454 prop_binCRWS_electron2018_bin14_ZZtoLep
455 prop_binCRWS_electron2016M_bin12_ZZtoLep
456 prop_binCRWS_electron2016M_bin9_ZZtoLep
457 prop_binCRWS_electron2016M_bin12_VBS_SSWW_SM
458 prop_binCRWS_electron2016M_bin5_ZZtoLep
459 prop_binCRWS_electron2017_bin11_ZZtoLep
460 prop_binCRWS_electron2017_bin6_TVX
461 prop_binCRWS_electron2017_bin6_Triboson
462 prop_binCRWS_electron2018_bin14_TVX
463 prop_binCRWS_electron2018_bin6_Triboson
464 prop_binCRWS_electron2018_bin8_Triboson
465 prop_binCRWS_muon2016M_bin10_Triboson
466 prop_binCRWS_muon2017_bin12_Triboson
467 prop_binCRWS_muon2017_bin12_ZZtoLep
468 prop_binCRWS_muon2017_bin7_Triboson
469 prop_binCRWS_muon2017_bin7_ZZtoLep
470 prop_binCRWS_muon2018_bin12_Triboson
471 prop_binCRWS_muon2018_bin5_Triboson
472 prop_binCRWS_muon2018_bin5_VG
473 FES
474 QCDScale_WZ
475 prop_binCRWS_electron2016M_bin10_VBS_SSWW_SM
476 prop_binCRWS_electron2016M_bin5_TVX
477 prop_binCRWS_electron2016M_bin5_Triboson
478 prop_binCRWS_electron2016M_bin5_VBS_SSWW_SM
479 prop_binCRWS_electron2016M_bin9_VBS_SSWW_SM
480 prop_binCRWS_electron2017_bin11_TVX
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
481 prop_binCRWS_electron2017_bin11_Triboson
482 prop_binCRWS_electron2017_bin11_VBS_SSWW_SM
483 prop_binCRWS_electron2017_bin13_VBS_SSWW_SM
484 prop_binCRWS_electron2017_bin13_ZZtoLep
485 prop_binCRWS_electron2017_bin6_ZZtoLep
486 prop_binCRWS_electron2018_bin14_VBS_SSWW_SM
487 prop_binCRWS_electron2018_bin6_TVX
488 prop_binCRWS_electron2018_bin6_VBS_SSWW_SM
489 prop_binCRWS_electron2018_bin6_ZZtoLep
490 prop_binCRWS_electron2018_bin8_TVX
491 prop_binCRWS_electron2018_bin8_VBS_SSWW_SM
492 prop_binCRWS_electron2018_bin8_ZZtoLep
493 prop_binCRWS_muon2016M_bin10_VBS_SSWW_SM
494 prop_binCRWS_muon2016M_bin10_ZZtoLep
495 prop_binCRWS_muon2017_bin12_VBS_SSWW_SM
496 prop_binCRWS_muon2017_bin7_VBS_SSWW_SM
497 prop_binCRWS_muon2018_bin12_VBS_SSWW_SM
498 prop_binCRWS_muon2018_bin12_ZZtoLep
499 prop_binCRWS_muon2018_bin5_TVX
500 prop_binCRWS_muon2018_bin5_VBS_SSWW_SM
501 prop_binCRWS_muon2018_bin5_ZZtoLep
502 tau_vsele_2016M
503 tau_vsele_2017
504 tau_vsele_2018
505 tau_vsmu_2016M
506 tau_vsmu_2017
507 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$